



FitGuard

AI-Powered Fitness and Nutrition Coach App

MOBILE COMPUTING PROJECT REPORT

Course Information

Course: Mobile Computing Lab

Instructor: Eng. Rania El-Sayed

Semester: Spring 2026 | 4th Year — Computer Engineering

Team Members

Team Members

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I. Introduction

FitGuard is an innovative, cross-platform mobile application developed using the Flutter framework, designed to revolutionize the home fitness and health management experience. The core objective of this project is to effectively bridge the gap between expensive professional personal coaching and self-directed, often risky, home workouts.

By leveraging advanced **Artificial Intelligence (AI) and computer vision**, FitGuard introduces a real-time pose estimation feature that analyzes the user's movements through their smartphone camera. It provides instant visual and auditory feedback to correct exercise posture, ensuring user safety and maximizing physical gains. Furthermore, the application features a dynamic, personalized nutrition planner tailored to individual goals and macros, alongside a comprehensive progress-tracking dashboard. FitGuard's mission is to provide an accessible, all-in-one smart fitness companion that guarantees safety, education, and results.

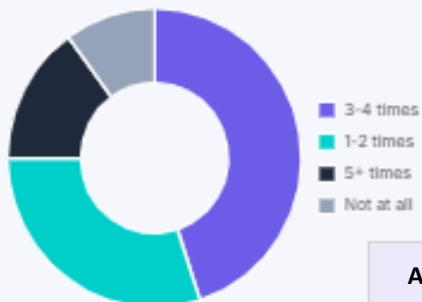
II. Section 01: UX Phase

a. Stage I: Research

1. Survey Analysis

To establish a data-driven foundation for our application, we conducted quantitative and qualitative research via an online survey. The survey targeted individuals engaged in home workouts and general fitness. It consisted of 10 structured questions designed to uncover current fitness habits, evaluate the acceptance of AI-driven real-time correction, and assess dietary planning challenges.

Q1: How many times a week do you usually work out?



- 3-4 times a week: **45%**
- Once or twice a week: **30%**
- 5 or more times a week: **15%**
- Not at all: **10%**

Analysis: The vast majority (90%) of our target audience engages in physical activity, indicating a high, consistent market demand for workout guidance and tracking.

Q2: Have you ever been injured or felt pain due to incorrect exercise form?



- No, but I worry about it: **45%**
- Yes, more than once / Yes, once: **35%**
- No, it does not concern me: **20%**

Analysis: A significant 80% of users have either experienced physical injury or actively fear it. This highlights a critical safety gap in solo workouts that FitGuard's AI correction directly

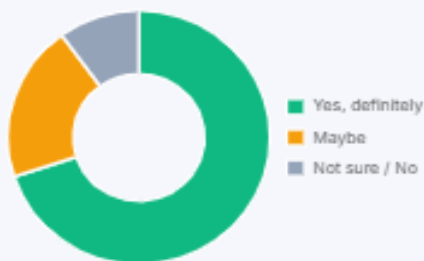
Q3: How do you currently make sure you are performing exercises correctly?



- Watch YouTube videos: **60%**
- I work out by instinct (No check): **20%**
- Record myself and review the video later: **15%**
- Ask a personal trainer: **5%**

Analysis: Reliance on static videos (60%) is high, yet it provides zero interactive feedback. Only 5% can afford or access a professional trainer.

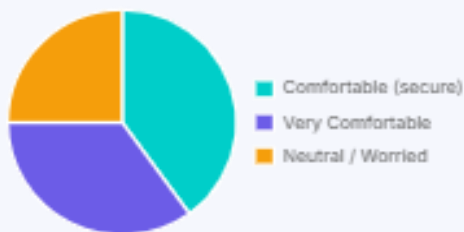
Q4: If an app used your phone camera to analyze your movements instantly, would you use it?



- Yes, definitely — this is exactly what I need: **70%**
- Maybe, if it is easy and seamless to use: **20%**
- Not sure / No: **10%**

Analysis: A 90% positive inclination proves massive product-market fit and user readiness for AI-integrated fitness solutions.

Q5: How comfortable are you with using your phone camera for AI analysis?



- Comfortable, provided data is secure: **40%**
- Very comfortable: **35%**
- Neutral or worried about privacy: **25%**

Analysis: While 75% are comfortable, the data emphasizes the necessity of implementing strict, transparent privacy policies regarding camera usage.

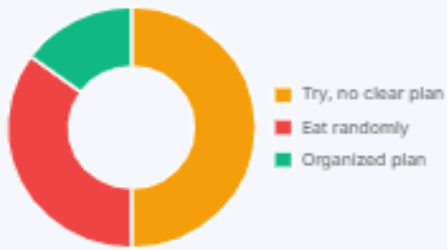
Q6: What type of feedback during a workout would be most useful to you?



- Instant voice alert / Text on screen: **55%**
- Highlighting the incorrect body part visually: **35%**
- Full summary after the workout finishes: **10%**

Analysis: Users strongly prefer instantaneous, actionable feedback (90%) over post-workout reports, validating the need for real-time processing.

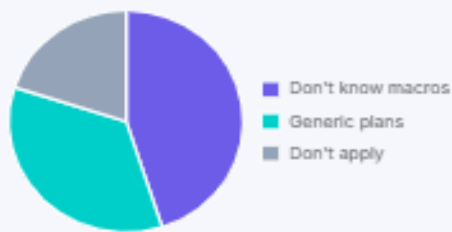
Q7: Are you currently following an organized nutrition plan?



- I try, but without a clear, consistent plan: **50%**
- No, I eat randomly: **35%**
- Yes, I follow a strict organized plan: **15%**

Analysis: 85% of users lack structured nutrition, highlighting a widespread difficulty in managing dietary habits alongside fitness.

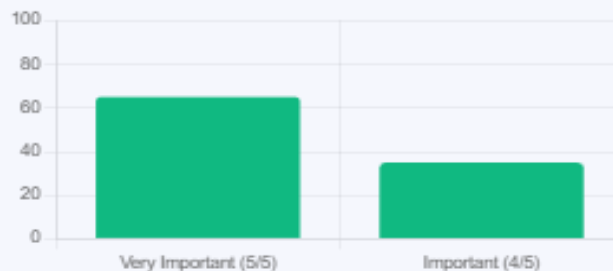
Q8: What is your primary challenge regarding sports nutrition?



- Ignorance of the right macros (Protein/Calories) for my body: **45%**
- Available plans on the internet are generic: **35%**
- I know the principles but lack the discipline to apply them: **20%**

Analysis: Personalization and macro-education are the largest hurdles, confirming the necessity of a smart nutrition calculator.

Q9: How important is a personalized nutrition feature to you?



- Highly Important (5/5): **65%**
- Important (4/5): **35%**

Analysis: A 100% consensus (rating 4 or 5) dictates that personalized nutrition is not an optional feature, but a mandatory requirement for user retention.

Q10: (Open Question) What is the most desired feature you want in an AI fitness app?

"Alert me immediately if my back arches during deadlifts," "I need an affordable alternative to gym trainers," and "Tell me exactly how many calories I need based on the workout I just finished."

Summary: The qualitative responses perfectly align with FitGuard's core value propositions: safety, affordability, and integrated nutritional guidance.

- Survey Link: <https://forms.gle/MRugbCWQVZRdCqTp7>

Conclusions

*Our comprehensive survey analysis reveals that the traditional self-directed home-workout model is fundamentally flawed. While users exhibit high motivation, they are paralyzed by the fear of injury and the prohibitive costs of professional coaching. Existing digital solutions, predominantly static video platforms, fail to provide the critical real-time interactive feedback users crave. Additionally, the disconnect between exercise and nutrition leaves users frustrated with generic dietary advice. The data unequivocally validates the core architecture of **FitGuard**: deploying AI-driven pose estimation to ensure immediate physical safety, coupled with a dynamic, highly personalized nutrition algorithm.*

Gains and Pain Points

User Pain Points

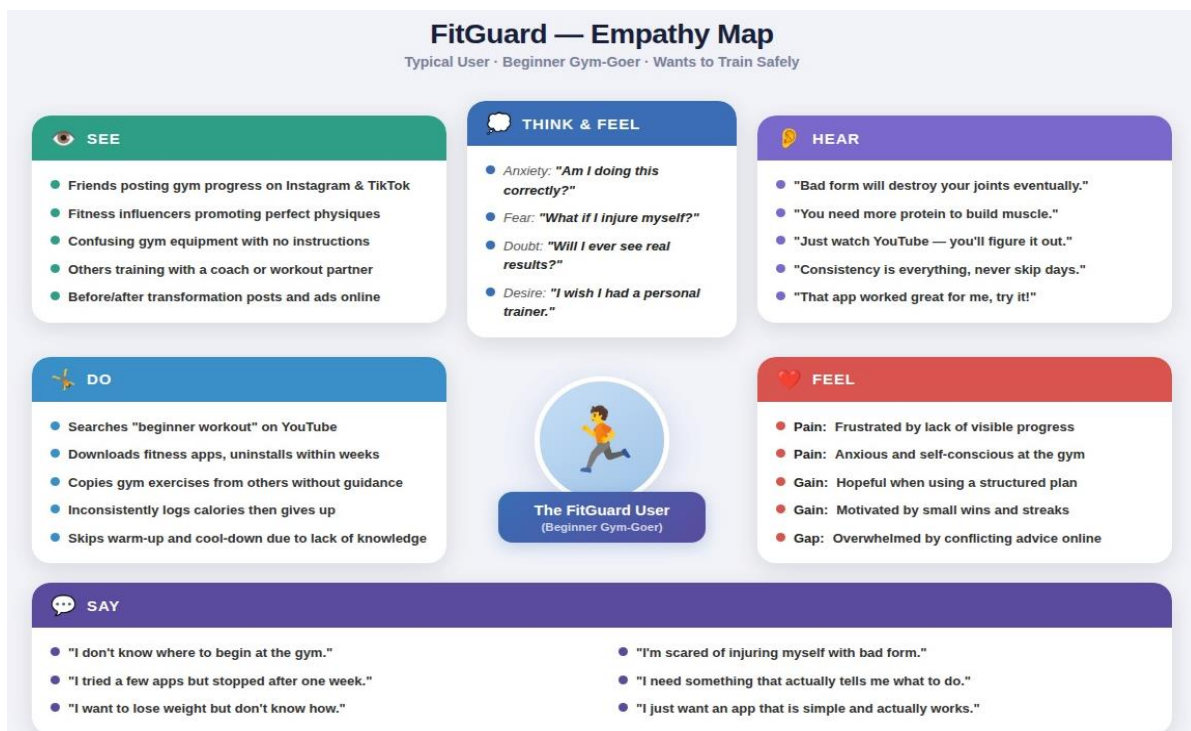
- **High Safety Risks:** Elevated anxiety and actual occurrence of musculoskeletal injuries due to improper posture.
- **Financial Barriers:** Professional personal training and customized dietitians are financially inaccessible.
- **Delayed Feedback Loop:** Recording videos for retrospective review fails to prevent mistakes dynamically *during* execution.
- **Nutritional Ambiguity:** Deep frustration with generic diets that ignore individual metabolic rates and goals.

Anticipated Gains

- **Real-Time Form Correction:** Receiving instantaneous visual and auditory cues to rectify posture safely.
- **Cost-Effective Democratization:** Accessing expert-level coaching directly through ubiquitous smartphone hardware.
- **Unified Ecosystem:** A centralized dashboard seamlessly integrating both biomechanical analysis and nutritional tracking.
- **Psychological Confidence:** Exercising with peace of mind, knowing an intelligent system is monitoring physical safety.

2. Empathy Map & Personas

A. Empathy Map: A synthesized visualization mapping the target user's emotional and cognitive state regarding fitness solutions.



B. User Personas

AHMED MOHAMED

PROFILE

Gender : Male

Age : 22

Education : Bachelor's degree (Engineering)

Occupation : University Student

Address : Cairo, Egypt

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BIOGRAPHY

Ahmed is a second-year engineering student who started working out 6 months ago. He trains at home and at the gym, relying on YouTube for guidance. He has no personal trainer due to cost and struggles with correct form and nutrition.

MOTIVATIONS

- Build muscle and improve his body shape

- Work out safely without getting injured

- Find an affordable alternative to a trainer

- See real progress week by week

GOALS

- Get AI-powered real-time form correction

- Follow a personalized nutrition plan

- Avoid injuries during workouts

- Track fitness progress consistently

FRUSTRATIONS

- Can't verify if his movements are correct

- Generic nutrition plans don't fit his needs

- Personal trainers are too expensive

- Had knee pain from incorrect squats

PERSONALITY

Introvert

Extrovert

Thinking

Feeling

Judging

Perceiving

Sensing

Intuition

TECHNOLOGY

Software

Social Media

Mobile App

BRANDS

YouTube · Instagram · MyFitnessPal

Khaled Hassan

AGE: 34

ROLE: Software Engineer

LEVEL: Intermediate

MAJOR: Computer Engineering

ABOUT

Khaled is a software engineer who works 8-8 hours a day at his leak. His work schedule is demanding and often leaves him mentally stained by the evening in his early inventites he was athletic. Ilut over the years he gained weight due to long meking hours 4e long working hours and lack of exericle. He wants to zagain his fitness but struggles to maintain consistency.

NEEDS

• Time-efficient workouts (30-45 minutes max)

• Automated calorie and macro tracking

• Visual progress analytics and charts

• Reminders and habit tracking

• Flexible workout plans when sessions are missed

GOALS

• Lose 15 kg and reduce belly fat

• Build a consistent 4-day workout routine

• Feel more energet

PERSONALITY

• Data-driven

• Analytical

• Results-focused

• Practical

Sara Ahmed

AGE: 21

ROLE: University Student

LEVEL: Beginner

MAJOR: Business Administration

ABOUT

Sara is a university student whu spends long heurs studying and ditending lectures jearex. She is axperion social media and enipus trying new theigle trence. She warks to stay healthy and fit but when struggles with time management between studying, social ins- and personal activities. Sara prefers simple rhin fitness routines that ean easily in into her daily schedule.

NEEDS

• Quick workouts (15-30 minutes)

• Beginner-friendly fitness guidance

• Simple calorie tracking

• Motivational reminders and achievements

• Feels overwhelmed by complex ftness plans

GOALS

• Stay active and maintain a health lifestyle

• Build a daily exercise habit

• Track progress easily

PERSONALITY

• Social

• Motivated by trends

• Curious

• Goal-oriented

b. Stage II: Define a problem

In this stage, we synthesize our research findings to clearly articulate the core problems our users face. This clarity ensures that our ideation and development efforts remain strictly user-centric.

1. Pain Points Categorization

Based on our empathy mapping and survey analysis, user frustrations can be systematically categorized into three primary domains:

Category	Identified Pain Points
Safety & Execution	- High risk of musculoskeletal injury due to improper form during unmonitored home workouts. - Total absence of immediate, actionable feedback to correct posture in real-time.
Financial & Access	- Prohibitive costs associated with hiring professional personal trainers or dietitians. - Inaccessibility to tailored, expert guidance for average or beginner users.
Nutritional Fragmentation	- Reliance on generic, "one-size-fits-all" diet plans that ignore individual metabolic rates. "App fatigue" caused by toggling between separate applications for fitness routines and macro counting.

2. 5Ws and H Analysis

The 5W1H framework helps us comprehensively understand the context of the problem before jumping to solutions.

Dimension	Description
Who	Fitness beginners, busy professionals, and home-workout enthusiasts who lack professional guidance.
What	They struggle with executing exercises safely, tracking their progress effectively, and managing personalized sports nutrition without external help.
Where	Primarily in home environments, living rooms, or unsupported gym settings.
When	During their daily workout routines, specifically when attempting new or compound exercises, and during meal-planning sessions.
Why	Due to the high cost of personal trainers, the static/non-interactive nature of online tutorials (e.g., YouTube), and the complexity of calculating accurate nutritional macros.
How	This current gap results in physical injuries, loss of motivation, inconsistent progress, and frustration with app fragmentation.

3. Problem Statements

Using the standard UX formula **[User] needs a way to [User Need] because [Insight]**, we defined the following core problem statements:

PROBLEM STATEMENT 1: THE SAFETY GAP

*"Home-workout beginners need a reliable, affordable method to **verify their exercise form in real-time**, because **relying on static video tutorials often leads to incorrect execution and potential musculoskeletal injuries.**"*

PROBLEM STATEMENT 2: THE NUTRITION DISCONNECT

*"Busy professionals need a streamlined, automated approach to **generate and track personalized nutrition plans**, because **generic diets and fragmented tracking apps disrupt their consistency and fail to meet their unique metabolic needs.**"*

PROBLEM STATEMENT 3: THE ACCESSIBILITY ISSUE

*"Budget-conscious fitness enthusiasts need an affordable alternative to **professional coaching**, because **the prohibitive costs of personal trainers leave them without the necessary expert guidance to achieve their goals safely.**"*

c. Stage III: Ideate a solution

1. Goal Statements

Translating our problem statements into actionable design goals using the template: **Our goal is to [Action] so that [User] can [Benefit]**.

Goal 1: Safety & AI

Our goal is to **implement a real-time, AI-driven pose estimation feature** so that **home-workout beginners can exercise safely with instant posture correction without risking injury.**

Goal 2: Unified Nutrition

Our goal is to **develop an integrated macronutrient calculator and meal planner** so that **busy professionals can maintain dietary consistency without the friction of using multiple apps.**

Goal 3: Accessibility

Our goal is to **create a cost-effective digital coaching mobile platform** so that **budget-conscious fitness enthusiasts can receive expert-level guidance anywhere, at any time.**

2. Proposed vs. Selected Solution

Proposed Solutions:

- **Solution A (IoT Wearable):** A physical smart-suit or wearable sensors that track joint movements and connect to a web dashboard for analysis.
- **Solution B (Video Subscription Platform):** A traditional web application offering pre-recorded, high-quality video tutorials and downloadable PDF generic diet plans.

- **Solution C (AI-Powered Mobile App):** A cross-platform mobile application that utilizes the smartphone's built-in camera and machine learning (Pose Estimation) to provide real-time form correction, combined with dynamic nutrition tracking.

Selected Solution: Solution C (FitGuard Mobile App)

Reason for Selection: Solution A (IoT) was rejected due to high manufacturing costs and low accessibility for average users. Solution B (Static Videos) was rejected because it fails to solve the core "real-time feedback" and "safety" pain points identified in our research. **Solution C was selected** because it leverages hardware the user already owns (a smartphone camera), making it highly accessible, exceptionally cost-effective, and technically capable of delivering the interactive, real-time AI feedback required to ensure user safety.

3. Competitor Analysis & Previous Approaches

To ensure FitGuard outperforms existing solutions, we analyzed major competitors to identify their weak points and define our competitive advantages.

Previous Approach (Competitor)	Identified Weak Points	FitGuard's Outperforming Features
YouTube / Static Video Apps (e.g., Nike Training Club)	Offers only one-way communication. Users must guess if they are replicating the movement correctly. No automatic rep counting or form correction.	AI Trainer Dashboard: Provides two-way interaction via an AI camera overlay that actively monitors joints, flags errors (e.g., "Good Depth"), and counts reps automatically.
MyFitnessPal / Cronometer	Solely focused on nutrition. Lacks native, guided workout integration, forcing users to manually estimate and log calories burned during workouts.	Unified Ecosystem: Calorie expenditure from AI-tracked workouts directly and automatically adjusts the daily nutritional macros within the same application.
Human Personal Trainers	Highly expensive, requires scheduling, and is geographically limited to a specific gym or location.	Accessibility & Affordability: Available 24/7 in the user's living room at a fraction of the cost, utilizing a digital AI Coach for guidance and recovery tips.

4. Project Brief & Feature Specification

The Problem Brief: Users attempting to work out at home suffer from a lack of affordable, real-time professional guidance, leading to poor form, injuries, and inconsistent progress. This is compounded by the frustration of managing fragmented nutrition apps.

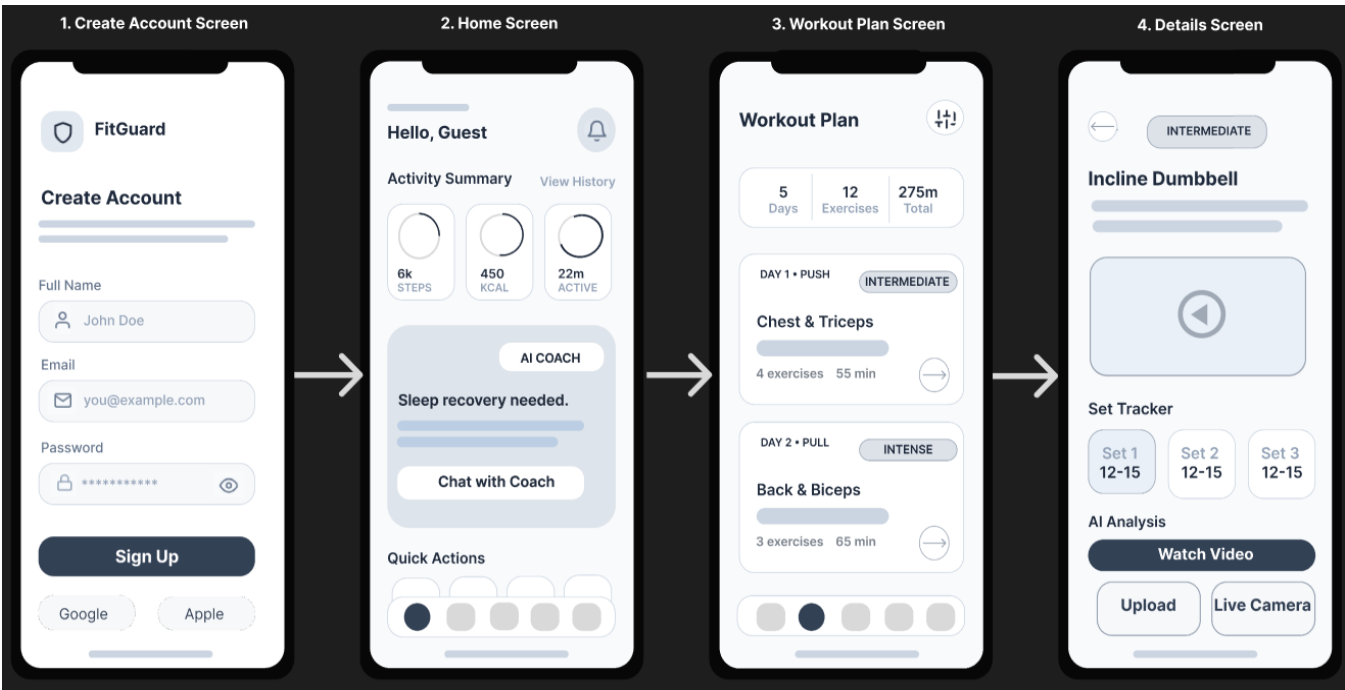
The Solution Brief: FitGuard is a smart, cross-platform mobile fitness companion that acts as a virtual personal trainer. It utilizes on-device Machine Learning to analyze the user's live camera feed, ensuring exercises are performed safely, while simultaneously automating customized meal planning.

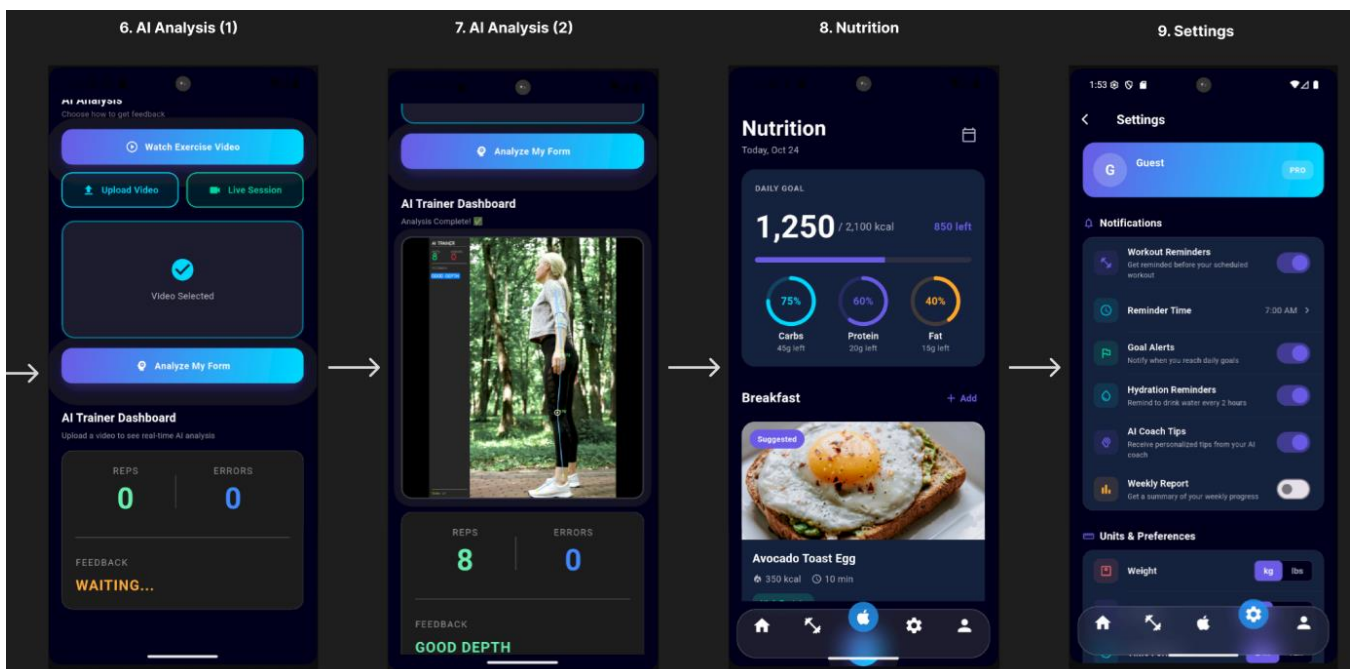
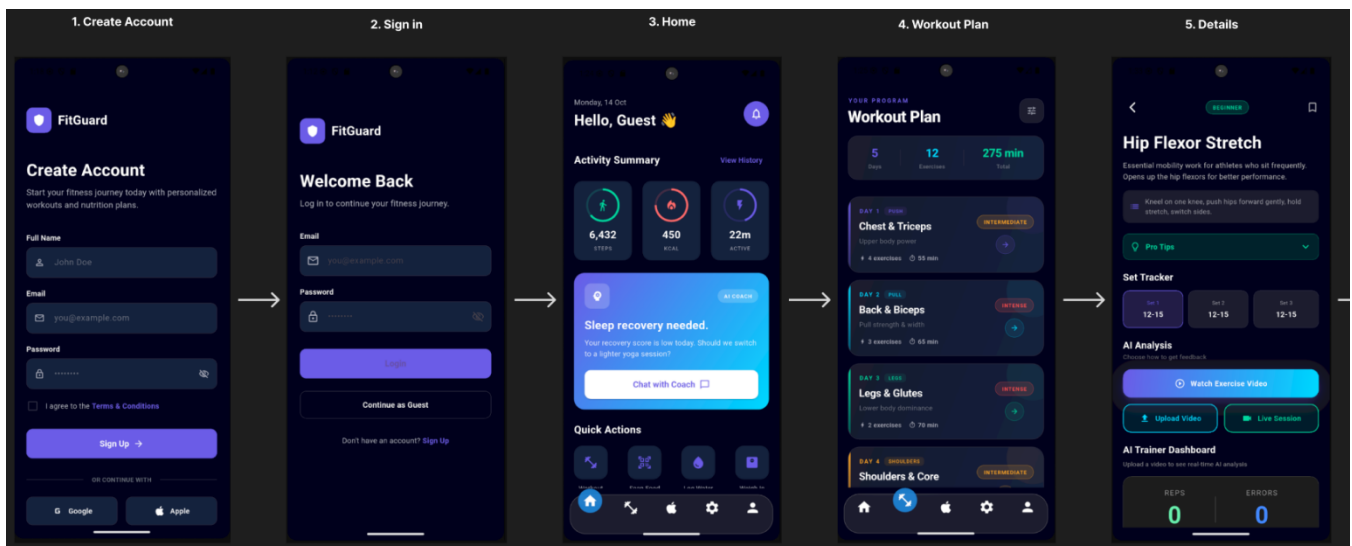
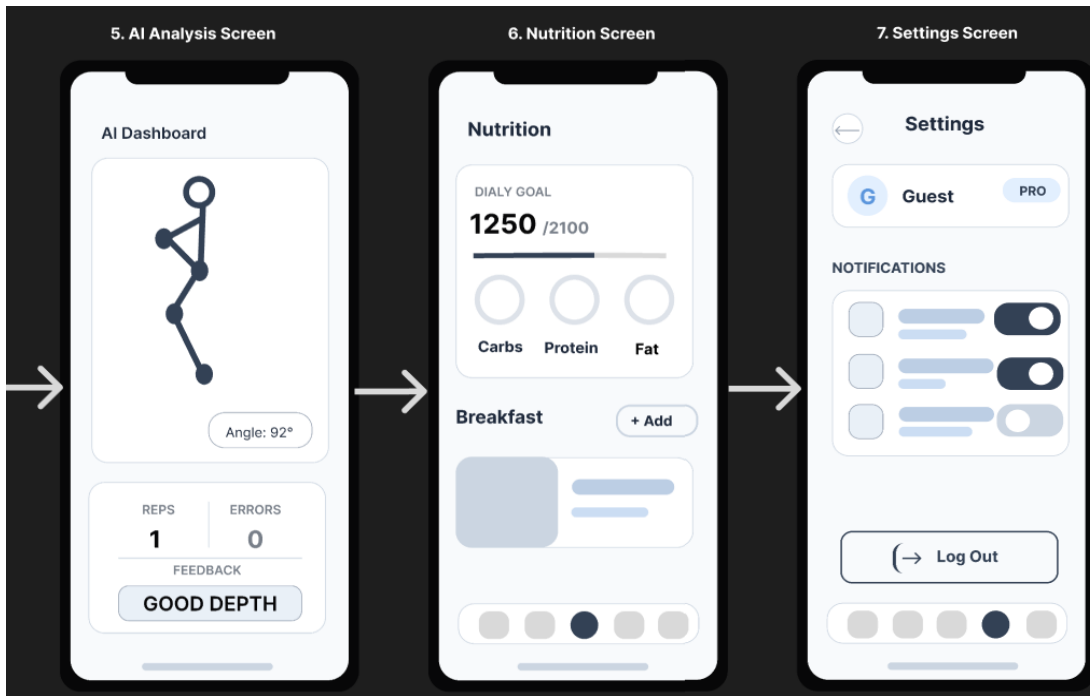
Proposed App Features:

- **Live AI Pose Estimation:** Real-time joint tracking and posture correction via the smartphone camera.
- **Automated Analytics:** Automatic rep counting and error logging during live sessions.
- **Personalized Macro Planner:** Dynamic calculation of daily Carbohydrates, Protein, and Fat requirements.
- **Categorized Workout Plans:** Structured routines (e.g., Push, Pull, Legs) tailored to user levels.
- **Interactive AI Coach:** A smart assistant providing recovery recommendations and daily tips.
- **Progress Dashboard:** Activity rings and charts to track steps, calories burned, and active minutes.

d. Stage IV: Wireframes & Site Map

Below is the low-fidelity structural blueprint of the application along with the navigational flow (Site Map), followed by the high-fidelity UI design.





Interactive Wireframe — Figma

7 Screens: Sign Up | Login | Home | Workouts | Nutrition | AI Analysis | Settings

https://www.figma.com/design/OSess0tz7Goj6efCH7eVDo/FitGuard_Wireframe?node-id=0-1&m=dev&t=fv59sbvtLKKhKp4R-1

III. Section 02: Frontend Design & Development

For the frontend development, we selected **Flutter** as our primary *Mobile Cross-Platform Framework*. Flutter was chosen for its ability to deliver a seamless, high-performance, native-like experience on both Android and iOS devices from a single codebase. The interface adopts a "Dark Mode First" philosophy with vibrant neon accents to reflect an energetic and modern fitness aesthetic.

1. Design Colors and Fonts

To ensure a consistent and visually appealing UI, we established a strict design system. Below is the visualization of our color palette and typography hierarchy.

FITGUARD COLOR PALETTE



Primary

#6C5DD3



Secondary

#00E5FF



Success

#10B981



Background

#0B0D17



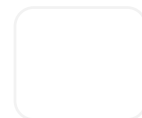
Surface

#151828



Warning/Error

#EF4444



Text Main

#FFFFFF

TYPOGRAPHY — INTER

The 'Inter' font family was chosen for its clean, geometric readability on mobile screens. Four distinct font roles manage the visual hierarchy.

Title

Inter Bold

24 - 32px

Main screen headers
(e.g., "Create Account",
"Hello, Guest").

Header

Inter SemiBold

16 - 20px

Section headers,
exercise names, and
card titles.

Regular

Inter Regular

12 - 14px

Body text, descriptions,
and standard form
inputs.

Accent

Inter Bold

10 - 12px

Button labels,
uppercase pill badges,
and input labels.

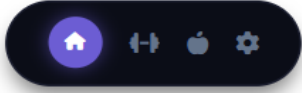
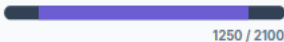
2. Custom Buttons

We developed custom, reusable button components in Flutter to guarantee interactive consistency.

Component Name	Style Details	Where it is used	Design Preview
Primary Solid Button	Solid Primary color (#6C5DD3) background, 12px border radius, bold white text.	Sign Up, Login, and primary submission forms.	
Gradient Action Button	Horizontal linear gradient (Primary Purple to Secondary Cyan), 12px radius, bold white text.	Major feature triggers like "Watch Exercise Video" and "Analyze My Form".	
Secondary Outline Button	Transparent background, 2px solid Secondary Cyan border, 12px border radius, Cyan icon and text.	"Upload Video", "Live Session" options in AI screen.	
Status Pill Badge	Dark surface background (#151828) with dynamic colored border and text (e.g., Orange for Intermediate, Red for Intense).	Workout Plan tags (e.g., INTERMEDIATE, INTENSE).	

3. Other Custom Components

To elevate the user experience, several custom UI components were implemented for data visualization and seamless navigation.

Component Name	Style Details	Where it is used	Design Preview
Floating Bottom Nav	Floating dark pill container (#0B0D17) with drop shadow. Active icon gets a glowing Primary color circular background.	Main app navigation across all primary screens.	
Macro Progress Bar	Horizontal thick bar with dual-tone fill (background dark slate, foreground Primary Purple).	Nutrition Dashboard (Calories consumed vs. Goal).	
AI Feedback Badge	High-contrast solid badge (Success Green or Error Red) overlaid on the live camera feed for immediate visibility.	AI Pose Estimation Camera Screen.	

4. Other Design Aspects

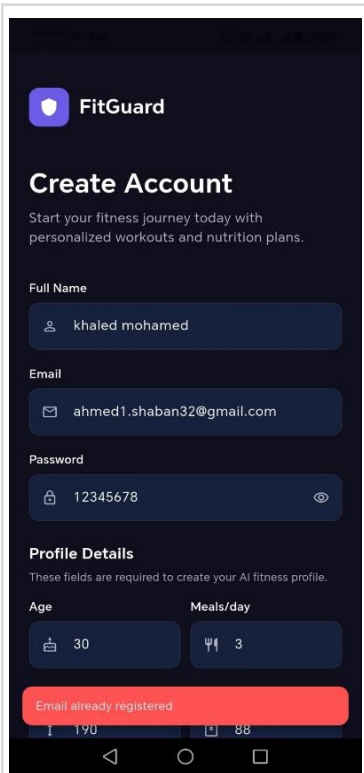
Modern UI Trends & Dark Mode First

The entire app utilizes a strict **Dark Theme architecture**. Dark mode is specifically chosen for fitness apps because it saves battery during long workout sessions, reduces glare in low-light gym environments, and ensures that neon accents (like activity rings, gradients, and AI feedback) remain highly legible and motivating.

Furthermore, we incorporated **Glassmorphism** (frosted glass effect using Flutter's `BackdropFilter` widget) on overlapping elements like the "Chat with Coach" card gradient. This provides a premium depth effect, separating the foreground action items from the dark background.

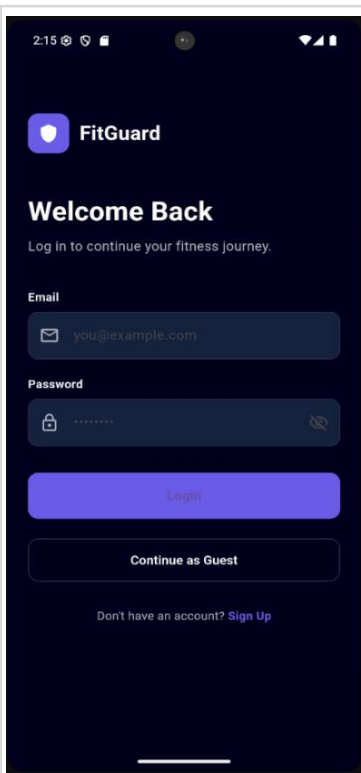
5. Screenshots for UI Design Screens

The following screenshots display the final implemented UI design of the core screens in FitGuard:



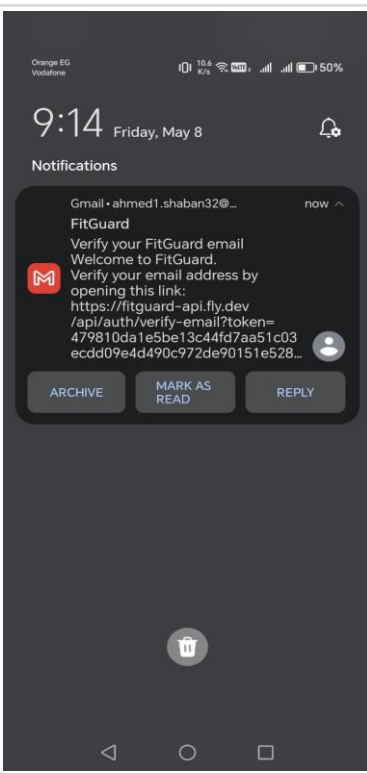
1. Create Account

User onboarding and registration interface with social login integration.



2. Sign In

Clean authentication screen allowing existing users to resume their journey.



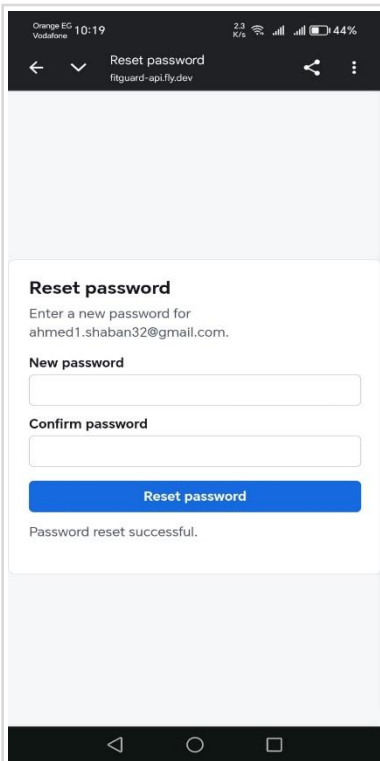
3. Email Verification

Real-time push notification for email verification ensuring secure user onboarding.



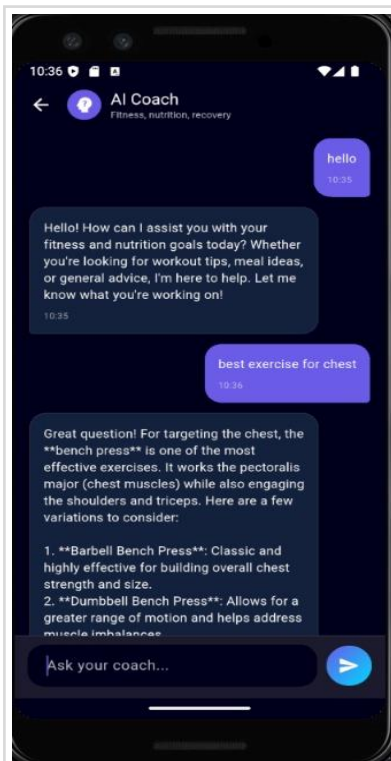
4. Forgot Password

Secure password recovery interface displaying dynamic feedback after link dispatch



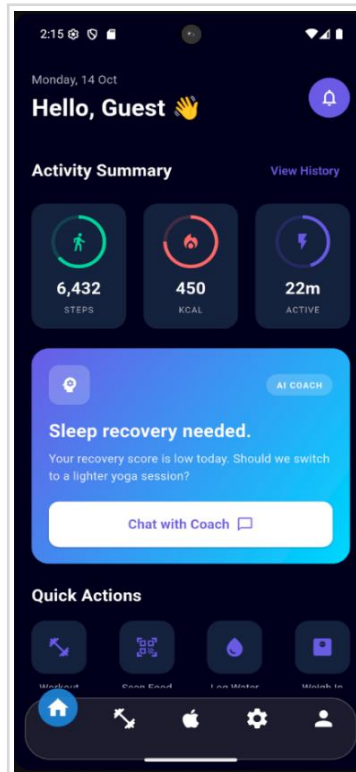
5. Password Reset

Secure and user-friendly interface for updating credentials and regaining account access seamlessly.



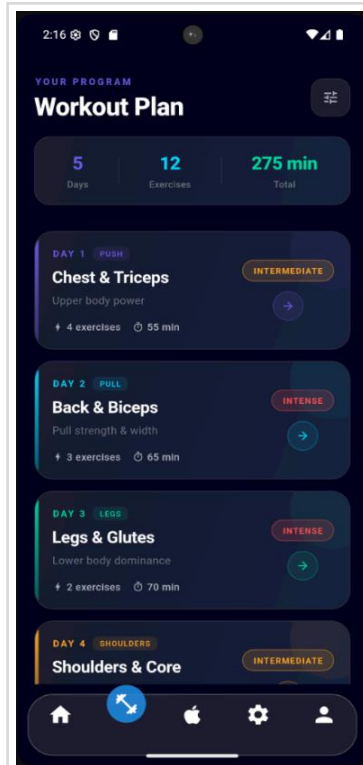
6. Interactive AI Coach

Dynamic conversational AI assistant delivering personalized fitness guidance, workout tips, and real-time support.



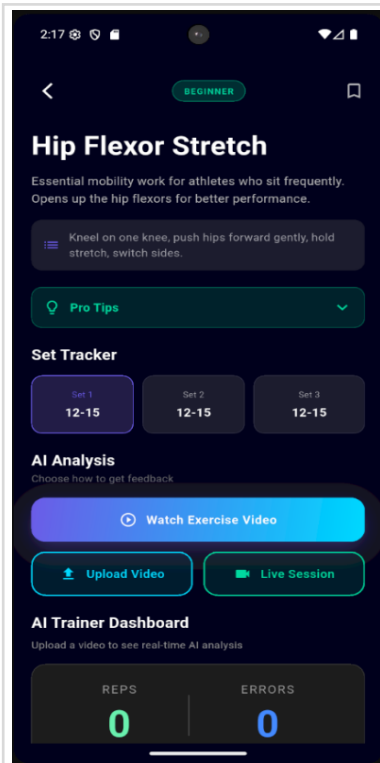
7. Home Dashboard

Displays daily activity rings, AI coach recommendations, and quick actions.



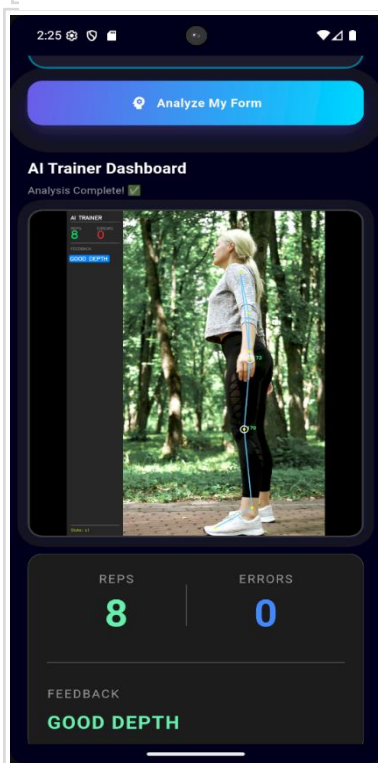
8. Workout Plan

Dynamic list of categorized exercises curated based on fitness level.



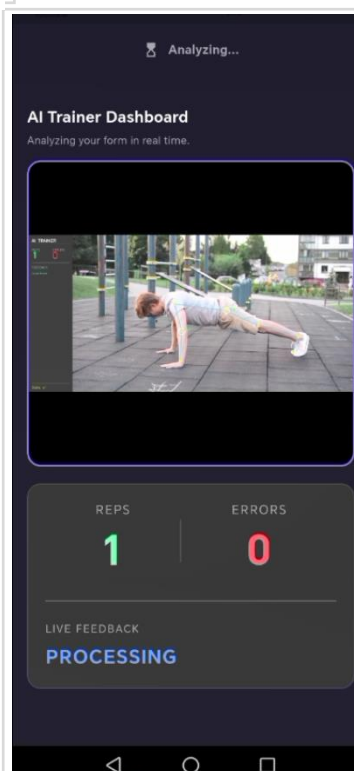
9. Exercise Details

Shows exercise instructions, set trackers, and triggers for the AI Camera integration.



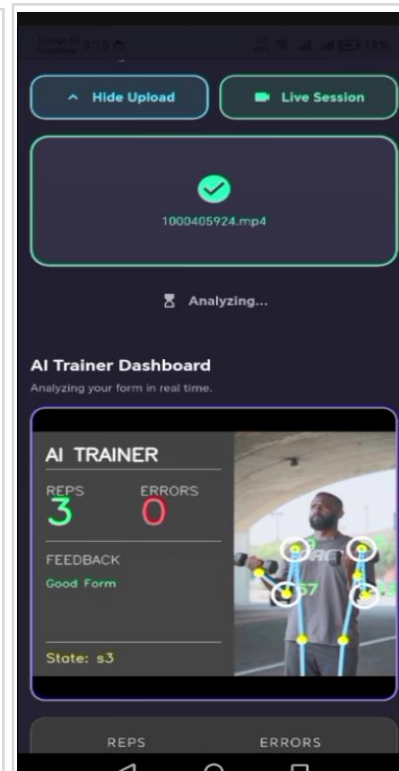
10. AI Analysis Core

Live camera feed with pose estimation skeleton, rep counter, and feedback.



11. AI Analysis Core

Live camera feed with pose estimation skeleton, rep counter, and feedback.



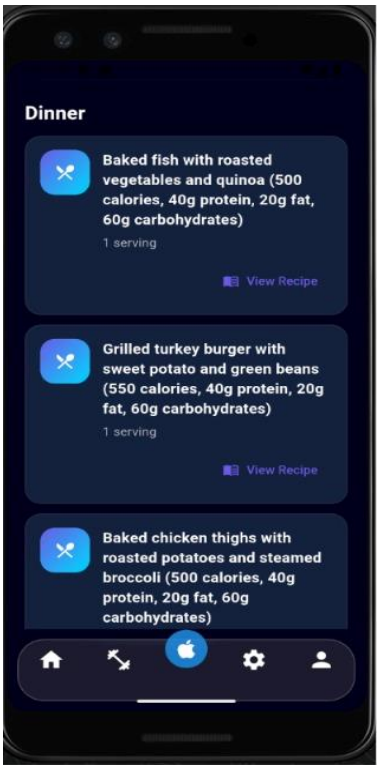
12. AI Analysis Core

Live camera feed with pose estimation skeleton, rep counter, and feedback.

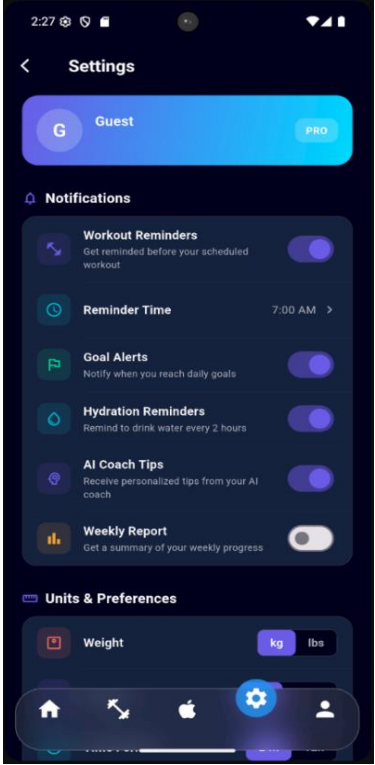


13. Nutrition Tracker

Visualizes daily caloric intake, macro distribution, and suggested meals.



14. Nutrition Tracker (2)



15. Settings Profile

Allows users to manage notification preferences and account details.

6. App Responsiveness

FitGuard is built utilizing Flutter's responsive layout guidelines. By implementing widgets such as MediaQuery, Flexible, and LayoutBuilder, we ensure that the UI scales flawlessly, adapting the grid columns and font sizes whether viewed on a compact smartphone or a larger tablet device.



▪ Technologies Stack

As requested, the following table specifies all the technologies integrated to develop both the Frontend and Backend architectures of FitGuard.

Category	Technology & Framework Used
Frontend Development	Flutter Framework (Dart) . Used to compile native cross-platform code for iOS and Android from a single codebase.
Backend Framework	Python & FastAPI . A modern, fast web framework for building APIs with Python, utilized for handling high-concurrency requests and AI integrations.
Database & ORM	PostgreSQL (Relational Database) accessed asynchronously via SQLAlchemy and <code>`asyncpg`</code> . Data validation is handled strictly by Pydantic models.
AI / Machine Learning	Computer Vision integration for real-time form error tracking. The backend receives dynamic <code>cv_metrics</code> and <code>form_errors</code> payloads to analyze performance.
UI/UX Design	Figma . Used for designing Wireframes, Site Maps, and high-fidelity prototypes.

IV. Section 03: Backend

The backend infrastructure of FitGuard is architected using **Python** and **FastAPI**, offering a robust, asynchronous RESTful API. We utilized **PostgreSQL** as our relational database, interfacing with it via **SQLAlchemy (AsyncSession)** to ensure high performance and data integrity across users' fitness and nutritional records.

1. Backend Architecture

Our project follows a clean modular structure to separate concerns. The backend directory is organized into several key layers:

Routers	Models	Schemas	Services
API endpoints (users.py, workouts.py) directing client requests.	SQLAlchemy ORM tables (user.py, workout.py, exercise.py).	Pydantic models ensuring strict data validation for inputs/outputs.	Business logic layer (plans.py) handling dynamic calculations.

2. Database Records & Snippets

FitGuard relies on a relational schema. Below are JSON representations (snippets) of the data rendered from our PostgreSQL database via Pydantic schemas in different parts of the app:

- **User Profile Record (users table)**

```
{ "id": 104, "full_name": "Ahmed Mohamed",  
  "gender": "male", "age": 22, "weight_kg":  
  75.5, "height_cm": 180.0, "activity_level":  
  "moderate", "primary_goal":  
  "muscle_building", "bmr": 1820.5 }
```

- **Workout Session (workout_sessions)**

```
{ "id": 5092, "user_id": 104,  
  "duration_seconds": 2700,  
  "total_target_reps": 40,  
  "total_actual_reps": 38, "improper_reps": 4,  
  "cv_summary": { "avg_back_angle": 92.4 } }
```

- **Dynamic Nutrition Plan (nutrition_plans table)**

```
{ "id": 214, "user_id": 104, "calorie_target": 2450, "macro_targets": { "protein_g": 135.9,  
  "carbs_g": 188.8, "fat_g": 60.4 }, "plan_data": { "primary_goal": "muscle_building", "source":  
  "estimated" } }
```

3. Dynamic Content Generation

FitGuard is not a static application; its core value lies in rendering Dynamic Content generated by intelligent backend services (services/plans.py). The API calculates and adjusts data dynamically based on individual user records before serving it to the Flutter frontend.

- **How Dynamic Content is Rendered:**

1. Algorithmic Nutrition Generation

When a user is created, the `generate_nutrition_plan_for_user` service dynamically calculates their `calorie_target` based on their BMR or the Harris-Benedict equation (using weight, height, age, and gender). Macros (Protein, Carbs, Fats) are dynamically multiplied based on their specific goal (e.g., adding 250 calories for Muscle Building).

2. Adaptive Workout Plans

The `generate_workout_plan_for_user` service checks if the user possesses advanced "InBody" metrics (body fat percentage, skeletal muscle mass). If present, the API dynamically elevates the plan's difficulty to "advanced". Otherwise, it renders a "foundational" plan, dynamically modifying the `plan_data` JSON object sent to the app.

3. Live AI Feedback & Session Logging

During a workout, the frontend streams AI computer vision metrics (CV metrics). The backend's `WorkoutSession` endpoint dynamically records `target_reps` vs. `actual_reps` and logs specific `form_errors` generated by the AI model. This dynamic data immediately renders on the user's progress dashboard in Flutter.